4

Fundamentals of Tooth Preparation

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Teeth require intervention (i.e., need some type of preparation) for various reasons: (1) caries lesion progression to the point that loss of tooth structure requires restoration; (2) tooth fracture compromising form and function with or without associated pain or sensitivity; (3) congenital malformation or improper position in need of reestablishment of form or function; (4) previous restoration with inadequate occlusal or proximal contact, defective (open) margins, or poor esthetics; or (5) as part of fulfilling other restorative needs (see Chapter 3). Every effort must be made to limit the risk of iatrogenic damage to adjacent tooth surfaces while seeking to prepare and restore caries lesions or defects. Teeth requiring intervention are prepared such that various restorative materials have the most predictable outcome.

This chapter defines tooth preparation and the historical classification of anatomic locations affected by caries lesions. Consideration is given to factors that directly impact preparation design, followed by description of the logic and procedural organization of preparation steps. Additional factors that must be considered in overall care of the patient may indirectly impact preparation design (Box 4.1; see Online Chapter 15). Chapters that are devoted to the preparation and restoration of specific lesions/defects elaborate on these additional factors. The ability to utilize the information presented here requires working knowledge of dental anatomy and a solid understanding of concepts presented in Chapters 1, 2, and 3.

In the past, most restorative treatment was for cavitated caries lesions, and the term *cavity* was used to describe a caries lesion that had progressed to the point that there was a breach in the surface integrity of the tooth. Likewise, when the affected tooth was treated, the cutting or preparation of the remaining tooth structure (to receive a restorative material) was referred to as *cavity preparation*. Currently, many indications for treatment are not related to carious destruction, and the preparation of the tooth no longer is referred to as *cavity preparation*, but as *tooth preparation*.

Tooth Preparation: Definition and Foundational Concepts

Tooth preparation is the mechanical alteration of a defective, injured, or diseased tooth such that placement of restorative material reestablishes normal form (and therefore function) including esthetic corrections, where indicated. Generally, the objectives of tooth preparation are to (1) conserve as much healthy tooth structure as possible, (2) remove all defects while simultaneously providing protection of the pulp–dentin complex, (3) form the tooth preparation so that, under the forces of mastication, the tooth or the restoration (or both) will not fracture and the restoration will not be displaced, and (4) allow for the esthetic placement of a restorative material where indicated.

G.V. Black presented a classification of tooth preparations according to diseased anatomic areas involved and by the associated type of treatment.[1] Black's classification originally was based on the observed frequency of caries lesions in various surface areas of teeth. These classifications were designated as Class I, Class II, Class III, Class IV, and Class V. Since Black's original classification, an additional class has been added, Class VI. Although the relative frequency of caries lesion locations may have changed over the years, the original classification is still used in the diagnosis of caries lesions (e.g., Class I Caries). In addition, the various classes are used to identify lesion-associated preparations and restorations (e.g., a Class I amalgam preparation or a Class I amalgam restoration).

All preparations required to treat pit-and-fissure caries are termed *Class I preparations*. These include preparations on (1) occlusal surfaces of premolars and molars, (2) occlusal two thirds of the facial and lingual surfaces of molars, and (3) the lingual surfaces of maxillary incisors. Preparations required to correct caries lesions that develop in the proximal surfaces of posterior teeth are termed *Class II preparations*. Preparations required to correct caries lesions that develop in the proximal surfaces of anterior teeth that do not include the incisal edge are termed *Class III preparations*. Preparations required to correct caries lesions or other defects that develop in the proximal surfaces of anterior teeth that include the incisal edge are termed *Class IV preparations*. Preparations required to correct caries lesions or other defects that develop in the gingival third of the facial or lingual surfaces of all teeth are termed *Class V preparations*. Preparations required to correct caries lesions or other defects that develop in the incisal edges of anterior teeth or the occlusal cusp tips of posterior teeth are termed *Class VI preparations*.

Much of the rationale supporting the development of tooth preparation techniques was introduced by Black.[1] Modifications of Black's principles of tooth preparation have resulted from the influence of Bronner, Markley, J. Sturdevant, Sockwell, and C. Sturdevant; from improvements in restorative materials, instruments, and techniques; and from the increased knowledge and application of preventive measures for caries.[2-6] Tooth preparation design takes into consideration the nature of the tooth (the structure of enamel, the structure of dentin, the position of the pulp in the pulp–dentin complex, the enamel connection to the dentin) *and* the nature of material to be used for restoration of the defect. For example, a diamond abrasive cutting instrument may be chosen to increase the roughness, and thereby surface area, of prepared walls and

• BOX 4.1 Factors to Consider Before and During Tooth Preparation

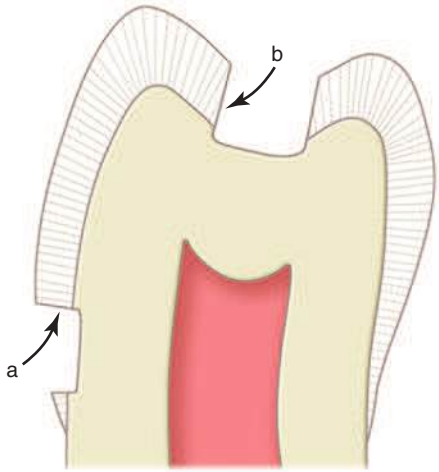
Patient Factors
- Desires
- Home care
- Risk status
- Age
- Cooperation
- Anesthesia

Lesion/Defect Factors
- Bone support
- Occlusion
- Severity
- Gingival status
- Pulpal status
- Fracture development

Anatomical Factors
- Enamel rod orientation
- Dentin thickness
- Pulp location
- Coronal contours
- Extent of previous restoration

Restorative Material Factors
- Physical properties
- Color characteristics
- Cost effectiveness

Procedural Factors
- Operator skill
- Instrument design
- Type of rotary cutting instrument
- Ability to isolate



• **Fig. 4.1** All enamel walls must consist of either full-length enamel rods on sound dentin (*a*) or full-length enamel rods on sound dentin supported on preparation side by shortened rods also on sound dentin (*b*).

margins when planning for an adhesively retained composite resin restoration (see Online Fig. 14.24).

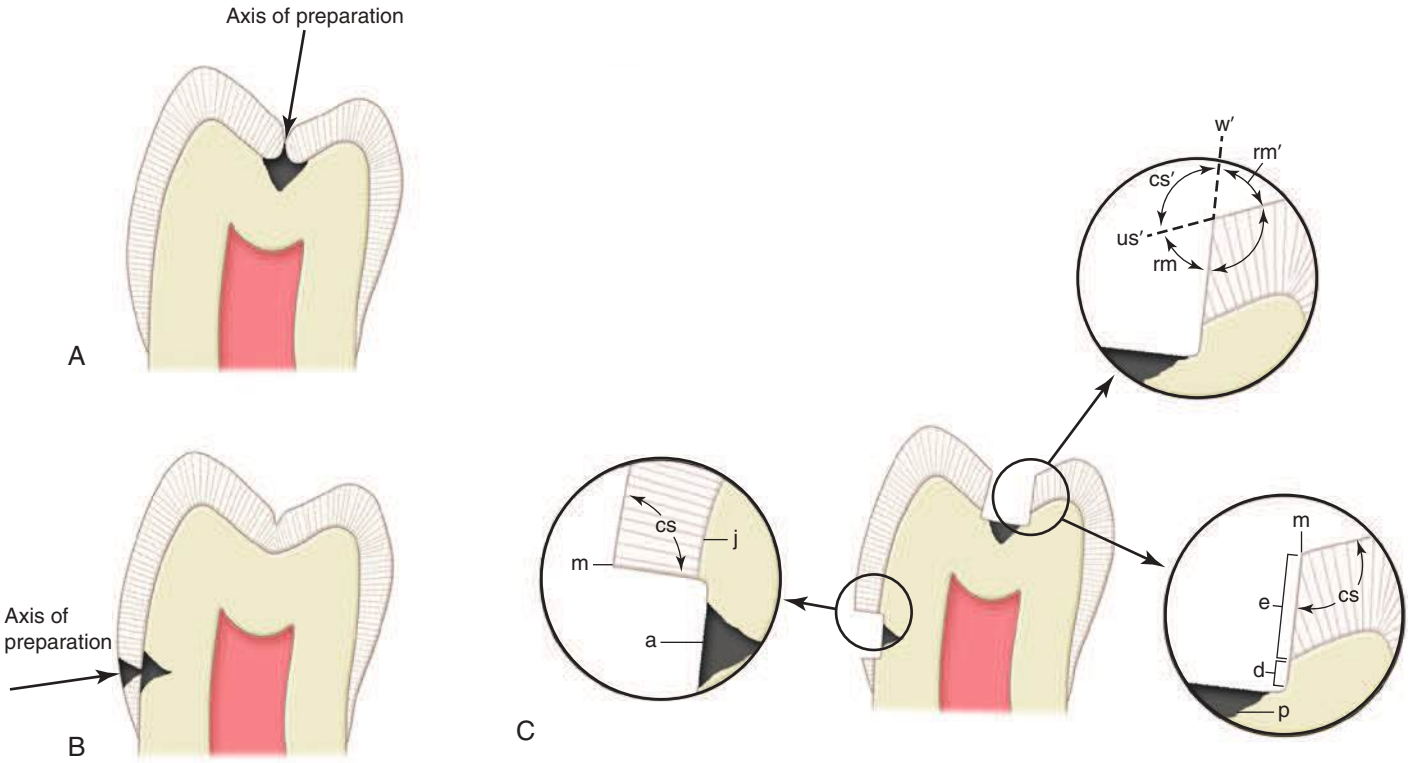
Highly mineralized enamel depends on the resiliency of its dentin support. Every preparation is designed to conserve as much dentin as possible for the strength of the enamel and the protection of the pulp. The durable attachment between enamel and dentin (the dentinoenamel junction [DEJ]) enables enamel to withstand the rigors of mastication. When caries (or any defect) has compromised the DEJ, then associated superficial enamel becomes prone to fracture under cyclic occlusal loading. Most preparation designs remove enamel that has lost its attachment to the underlying dentin (see Chapter 8 for exceptions). The nature of enamel formation (see Chapter 1) requires that the preparation walls be, at minimum, oriented 90 degrees to the external surface of the enamel so as to maintain a continuous connection with the essential supporting dentin beneath (Fig. 4.1, *a*). An enamel wall with this configuration is able to withstand the forces associated with occlusal loading. An even more durable wall configuration results when the preparation has full-length enamel rods buttressed by shorter enamel rods on the preparation side of the wall (Fig. 4.1, *b*). Tooth preparations must also include design features that take into account the physical limitations of the planned restorative material.

Dental restorative materials are best considered in terms of their ability to survive the stresses of the oral environment in comparison with natural tooth structure. Natural tooth structure is able to withstand the cyclic loading of mastication because of its ability to undergo small amounts of flexure without fracture or separation of the enamel from the dentin. Ideal restorative materials would be able to mimic the durability of natural tooth structure. All current restorative materials fall short of the ideal. Tooth preparation design takes into consideration that dental restorative materials are structurally either polycrystalline or polymeric.

Restorative materials that are *polycrystalline* in nature (e.g., dental amalgam, glass-ceramic) have very limited ability to flex without fracturing. Therefore they are prone to fracture when occlusal loading causes material flexure. Preparations for polycrystalline materials require removal of diseased tooth structure followed by a design that allows enough material thickness such that flexure under occlusal load will not occur. This process usually results in a preparation with fairly uniform (at least uniformly minimal) depths. Polycrystalline materials generally require a minimum thickness of 1.5 to 2.0 mm so as to withstand occlusal loading without flexure. The periphery of preparations for polycrystalline materials are designed to allow thickness (i.e., bulk) of the margins (edges) of the planned restoration. The design of the cavosurface margins for these materials is therefore as close to 90 degrees as possible as this marginal configuration allows maximum thickness of the polycrystalline material that will subsequently be placed in the preparation (Fig. 4.2; also see Fig. 1.12). Preparations for polycrystalline restorative materials often require strategic, additional removal of healthy tooth structure to allow for material limitations.

Restorative materials that are *polymeric* in nature (e.g., composite resin) have greater ability to flex without fracture. Therefore they are not prone to fracture when occlusal loading causes material flexure. Preparations for polymeric restoratives generally only require removal of the diseased tooth structure as these materials have no minimum material thickness requirement. The thin restoration will flex as needed. The periphery of preparations for polymeric materials do not require any particular design to allow for bulk of material at the margins of the planned restoration. Polymeric restorative materials may be as thin as is required to replace lost tooth structure and reestablish normal anatomy. For this reason, preparations for polymeric restorative materials generally allow maximum conservation of natural tooth structure and therefore are considered to be "minimally invasive" by design. The attachment between polymeric materials and enamel remains stable over time. However, the attachment between polymeric materials and dentin deteriorates over time.

Tooth preparations are usually limited to the clinical crowns of teeth. The *clinical crown* is the portion of the tooth (usually predominantly covered by enamel) that is exposed in the oral cavity. Frequently tooth preparation leaves much of the clinical crown surface uninvolved and is referred to as an *intracoronal tooth preparation.* The appearance of the completed preparation has been conceptually described as "boxlike" (Fig. 4.3). The goal of the operative dentist is always maximum conservation of any remaining



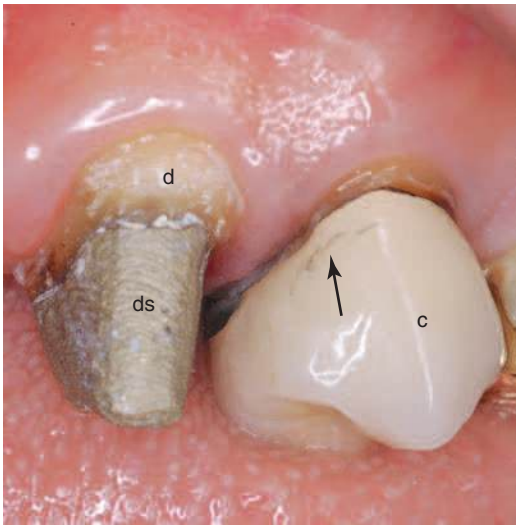
• **Fig. 4.2** Diagram of caries lesion development in the occlusal pit/fissure area (ICDAS 4) of a tooth (A) and in the smooth surface area on the facial (B). Sectional view (C) of initial stage of tooth preparations for lesions in A and B when planning for a polycrystalline restorative material such as amalgam. C, The preparation cavosurface angle (*cs*), axial wall (*a*), pulpal wall (floor) (*p*), enamel wall (*e*), dentinal wall (*d*), preparation margin (*m*), and DEJ (*j*). Note, in the upper exploded view, that the cavosurface angle (*cs*) may be visualized by imaginary projections of the preparation wall (*w′*) and of the unprepared surface (*us′*) contiguous with the margin, forming angle *cs′*. Angles *cs* and *cs′* are equal because opposite angles formed at the intersection of two straight lines are equal. Likewise, minimal restorative material angle *rm* is equal to angle *rm′*. Polycrystalline restorative materials require *rm* to approach 90 degrees. Note the axis of preparation aligned with the long axis of the mandibular posterior tooth crown.



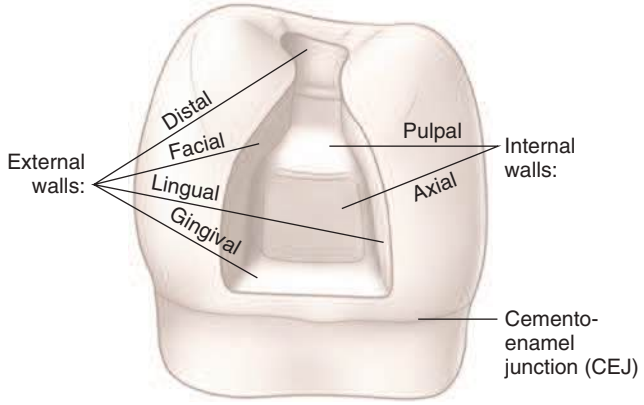
• **Fig. 4.3** Intracoronal preparation with "boxlike" appearance.

clinical crown knowing that the tooth has already been weakened from the carious loss of tooth structure. Controlled, conservative removal of any remaining tooth structure, based on the needs of the restorative material, is always accomplished with the awareness that the intracoronal restoration will *not* add strength to the tooth over the long term, regardless of the nature of the restorative material being used.

When carious destruction of the clinical crown is severe (i.e., the remaining enamel has lost a large amount of the dentinal support), additional efforts to encircle and reinforce the remaining tooth structure are required. These additional preparation efforts most frequently require removal of most or all of the remaining enamel and therefore include the whole *anatomic crown.* All external enamel surfaces are involved and the preparation effort is therefore referred to as an *extracoronal preparation.* In concept, all the enamel (at least the correct physical dimensions and frequently the physical appearance) is to be replaced. In addition, missing dentin may need to be replaced with an appropriate restorative material to act as a dentin substitute. The dentin substitute, along with remaining healthy dentin, acts to support the new restorative material that is replacing the enamel. The goal of the extracoronal preparation is to create enough physical space for the planned restorative material to restore the natural anatomy of the affected tooth. In general, the appearance of a completed extracoronal preparation is reminiscent of a tree stump and is referred to as "stumplike" (Fig. 4.4). The actual amount of space required depends directly on the physical properties of the restorative material to be used. The extracoronal restoration generally reestablishes the anatomy of the crown of the tooth (clinical or anatomic crown, depending on whether any enamel is remaining) and is therefore termed a "crown." The crown must extend well



• **Fig. 4.4** Extracoronal "stumplike" tooth preparation with dentin (*d*) and dentin substitute (core component of a cast post and core, *ds*). The adjacent tooth has been restored with a full porcelain-fused-to-metal crown (*c*). Note staining that has subsequently developed in areas of iatrogenic damage *(arrow)*.



• **Fig. 4.5** The external and internal walls (floors) for Class II tooth preparation required to treat occlusal and mesioproximal caries lesions.

beyond any dentin substitute (i.e., include remaining adjacent healthy tooth structure) if the restorative process is to successfully reestablish the strength required for durable function of the restored tooth. The more extensive the preparation, the greater the risk of iatrogenic damage of adjacent structures or restorations during procedures. Further information relative to extracoronal tooth preparations and restorations may be identified in textbooks devoted to this subject.

Dentistry has developed terminology useful in the communication of all aspects of preparation design and associated procedures. Tooth preparation terminology effectively describes preparation aspects with regard to complexity, anatomic location, three-dimensional orientation, and geometry.

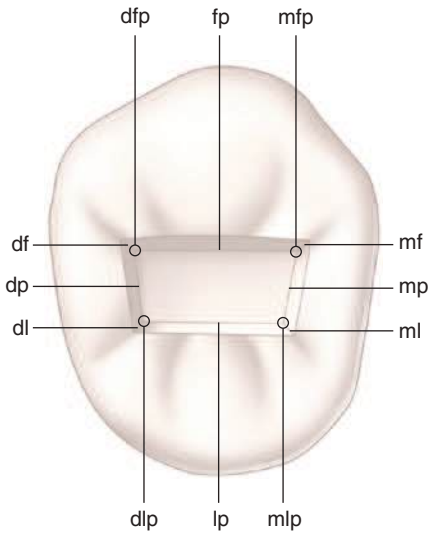
Tooth Preparation: Terminology

A tooth preparation is termed *simple* if only one tooth surface is involved, *compound* if two or three surfaces are involved, and *complex* if a preparation involves four or more surfaces. A preparation takes the name of the involved tooth surface(s)—for example, a defect on the occlusal surface is treated with an occlusal preparation. When discussing or writing a term denoting a combination of two or more surfaces, the *-al* ending of the prefix word is changed to an *–o.* For example, the angle formed by the lingual and incisal surfaces of an anterior tooth would be termed *linguoincisal line angle* and the tooth preparation involving the mesial and occlusal surfaces is termed *mesioocclusal preparation*. The preparation involving the mesial, occlusal, and distal surfaces is a *mesioocclusodistal preparation*. For brevity in records and communication, the description of a tooth preparation is abbreviated by using the first letter, capitalized, of each tooth surface involved. Examples are as follows: (1) A simple tooth preparation involving an occlusal surface is an "O"; (2) a compound preparation involving the mesial and occlusal surfaces is an "MO"; and (3) a complex preparation involving the mesial, occlusal, distal, and lingual surfaces is an "MODL."

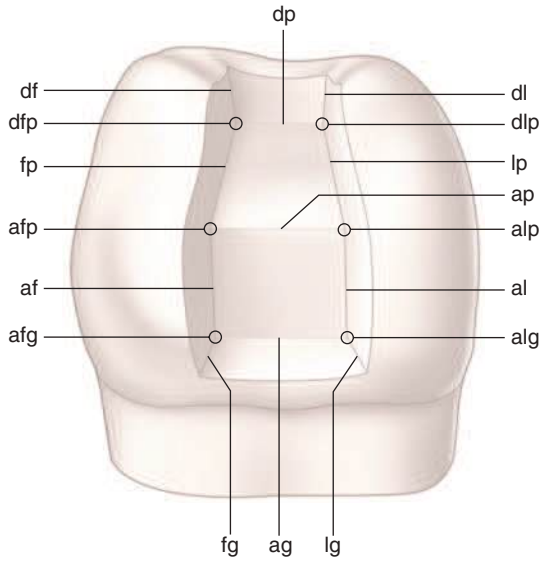
The process of creating a preparation in a tooth results in the formation of preparation walls or floors (Fig. 4.5). An *internal wall* is a prepared surface that does not extend to the external tooth surface. There are two types of internal walls. The *axial wall* is an internal wall that is oriented parallel to the long axis of the tooth. The *pulpal wall* is an internal wall that is oriented perpendicular to the long axis of the tooth and is located occlusal to the pulp. This internal wall may also be referred to as the *pulpal floor.* An *external wall* is a prepared surface that extends to the external tooth surface. Such a wall takes the name of the tooth surface (or aspect) that the wall is adjacent to (Fig. 4.5). The external wall that is approximately horizontal (i.e., perpendicular to the occlusal forces that are directed occlusogingivally and generally parallel to the long axis of the tooth crown) may also be referred to as a preparation *floor* (e.g., a gingival floor; see Fig. 4.5). The *enamel wall* is that portion of a prepared external wall consisting of enamel (see Fig. 4.2C). The *dentinal wall* is that portion of a prepared external wall consisting of dentin, in which mechanical retention features may be located (see Fig. 4.2C). Tooth preparation features or sections that are parallel (or nearly so) to the long axis of the tooth crown are commonly described as *vertical*, such as vertical height of cusps, or vertical walls. The term *longitudinal* may be used in lieu of *vertical*. Tooth preparation features that are perpendicular (or nearly so) to the long axis of the tooth are termed *horizontal* or *transverse*.

The junction of two or more prepared surfaces is referred to as the *angle.* This transition area from one surface to another is designed to be smooth and rounded, rather than abrupt or sharp, to limit stress concentration. A *line angle* is the junction of two planar surfaces of different orientation along a line (Figs. 4.6 and 4.7). An *internal* line angle is the line angle whose apex points into the tooth. The *external* line angle is the line angle whose apex points away from the tooth. The *point angle* is the junction of three planar surfaces of different orientation (see Figs. 4.6 and 4.7).

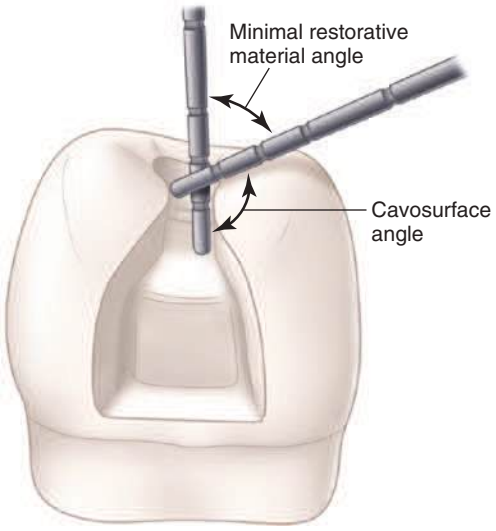
The *cavosurface angle* is the angle of tooth structure formed by the junction of a prepared wall and the external surface of the tooth. The actual junction is referred to as *cavosurface margin*. The cavosurface angle may differ with the location on the tooth, the direction of the enamel rods on the prepared wall, or the type of restorative material to be used. In Fig. 4.2C, the cavosurface angle *(cs)* is determined by projecting the prepared wall in an imaginary line *(w′)* and the unprepared enamel surface in an imaginary line *(us′)* and noting the angle *(cs′)* opposite to the cavosurface angle *(cs).* For better visualization, these imaginary projections may be formed by using two periodontal probes, one lying on the unprepared surface and the other on the prepared external tooth wall (Fig. 4.8).



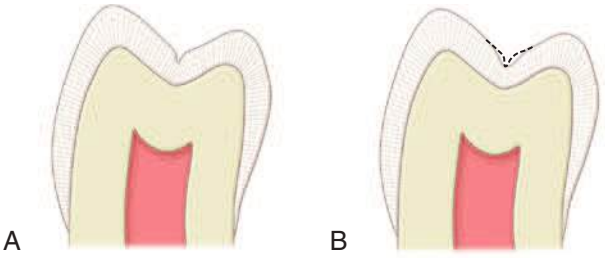
• **Fig. 4.6** Schematic representation (for descriptive purpose) of a Class I tooth preparation illustrating line angles and point angles. Line angles are faciopulpal (*fp*), distofacial (*df*), distopulpal (*dp*), distolingual (*dl*), linguopulpal (*lp*), mesiolingual (*ml*), mesiopulpal (*mp*), and mesiofacial (*mf*). Point angles are distofaciopulpal (*dfp*), distolinguopulpal (*dlp*), mesiolinguopulpal (*mlp*), and mesiofaciopulpal (*mfp*).



• **Fig. 4.7** Schematic representation (for descriptive purpose) of a Class II tooth preparation illustrating line angles and point angles. Line angles are distofacial (*df*), faciopulpal (*fp*), axiofacial (*af*), faciogingival (*fg*), axiogingival (*ag*), linguogingival (*lg*), axiolingual (*al*), axiopulpal (*ap*), linguopulpal (*lp*), distolingual (*dl*), and distopulpal (*dp*). Point angles are distofaciopulpal (*dfp*), axiofaciopulpal (*afp*), axiofaciogingival (*afg*), axiolinguogingival (*alg*), axiolinguopulpal (*alp*), and distolinguopulpal (*dlp*).



• **Fig. 4.8** Visualization of the cavosurface angle and the associated minimal restorative material angle for a typical amalgam tooth preparation.



• **Fig. 4.9** A, Enameloplasty on area of imperfect coalescence of enamel. B, No more than one third of the enamel thickness should be removed.

Tooth Preparation: Stages and Procedural Steps

Overview

It is imperative that the end result (i.e., the overall shape/goals of the preparation procedure) be envisioned/considered *before* the initiation of any step. The process of tooth preparation is conceptually divided into initial and final stages, each with several steps, so as to facilitate this mental discipline.

The initial stage of the preparation involves what is essentially a superficial surgical incision (with rotary instrumentation) into and through the enamel caries lesion to the depth of the DEJ followed by lateral extension of the preparation walls, at this limited depth, so as to fully expose the carious dentin lesion or defect. The lateral extension is controlled so as to only remove enamel with a compromised (demineralized) DEJ. The placement and orientation of the preparation walls are designed to resist fracture of the tooth or restorative material from masticatory forces principally directed parallel to the long axis of the tooth and to retain the restorative material in the tooth (except for a Class V preparation and Class III preparations with no component involving the occlusion).

Occasionally, very narrow grooves or fossae (that do not penetrate to any great depth into enamel) at the periphery of the preparation prevent the creation of preparation margins that are clearly defined and easily restored. Extension of the preparation to include these nondiseased irregularities would result in unnecessary removal of healthy tooth structure. A reasonable compromise may be to make a minor modification of the external enamel contours, in this peripheral area only, by selective removal of the surface enamel associated with the shallow, narrow developmental groove or fossa. The objective of this process, referred to as *enameloplasty*, is to create a smooth, saucer-shaped external surface that is self-cleansing or easily cleaned (Fig. 4.9 and Box 4.2).

• BOX 4.2 Enameloplasty and Prophylactic Odontotomy

Historically, enameloplasty was utilized as an ultraconservative procedure on the occlusal surfaces, which were deemed to be at risk of the development of a pit or fissure caries lesion. Extreme prudence was exercised in the selection of these areas and in the depth of enamel removed. This procedure was never used unless the area could be transformed into a cleansable groove (or fossa) by a minimal reduction of enamel, and unless occlusal contacts could be maintained. This procedure technically included a preparation stage but no restoration stage. Currently, clinical situations such as these (ICDAS 1 or 2) are managed by treatment with fluoride or placement of sealants. Research studies support the filling of fissures/pits and narrow grooves/fossae (i.e., "sealing") with low viscosity composite resin materials, without any mechanical alteration (enameloplasty) of the at-risk tooth anatomy.[9] Additionally, "prophylactic odontotomy" procedures were used in the past. These more aggressive procedures involved preparing developmental or structural imperfections of the enamel that were thought to be at increased risk of caries and filling the preparation with amalgam to prevent caries from developing in these sites. Prophylactic odontotomy is no longer advocated as a preventive measure.[42]

• BOX 4.3 Steps of Tooth Preparation

Initial Tooth Preparation Stage

Step 1: Initial depth and outline form
Step 2: Primary resistance form
Step 3: Primary retention form
Step 4: Convenience form

Final Tooth Preparation Stage

Step 5: Removal of defective restorative material and/or soft dentin
Step 6: Pulp protection
Step 7: Secondary resistance and retention forms
Step 8: External wall finishing
Step 9: Final procedures: debridement and inspection

Enameloplasty is accomplished as part of the initial preparation stage but does not involve extension of the preparation outline form and may be useful when creating a preparation to be restored with amalgam or glass-ceramic. Restorative material will not be placed into the recontoured area. The only difference in the restoration is that the thickness of the restorative material, at the enameloplastied margin, is slightly decreased because the pulpal depth of the preparation external wall is slightly decreased. This approach differs from including adjacent faulty (decalcified, discolored, poorly contoured) enamel areas, during preparation steps for composite restorations, as these defective areas are physically covered with adhesively bonded composite material as part of the restoration. Care is taken when choosing the area that will benefit from enameloplasty. Usually, a narrow groove should be included in the initial preparation extension if it penetrates to more than one third the thickness of the enamel in the area. If one third or less of the enamel depth is involved, the narrow groove may be removed by enameloplasty, thus limiting further extension of the tooth preparation. This procedure is also applicable to supplemental narrow grooves extending up cusp inclines. If the ends of these grooves were to be included in the tooth preparation, the cusp may be weakened to the extent that it would need to be reduced and covered with restorative material. Another instance in which enameloplasty may be indicated is the presence of a narrow groove that approaches or crosses a lingual or facial ridge. Inclusion of this narrow groove in the preparation would result in the involvement of two surfaces of the tooth, and use of the enameloplasty procedure may often limit the tooth preparation to one surface. Once the initial stage is completed, the final stage of preparation design may be accomplished.

The final stage is focused on (1) accurate management of the lesion/defect that has been isolated, (2) optimal protection of remaining tooth structure, and (3) preparation enhancements consistent with best long-term prognosis (durability) of the restoration. The goals of each step in the preparation stages must be thoroughly understood, and each step must be accomplished as precisely as possible if optimal treatment outcomes are to be obtained. The stages and steps in tooth preparation are listed in Box 4.3. The sequence of these steps may need to be altered when extensive caries has increased the risk of pulpal involvement (see Chapter 2).

The concepts of initial and final stages of tooth preparation are utilized for caries lesions that have progressed into dentin, have compromised the dentinal support of enamel, and therefore require surgical intervention. The information presented is comprehensive and specific primarily for tooth preparations designed to receive direct restorative materials that are not adhesively attached to the tooth structure and are polycrystalline in nature (i.e., amalgam). Major differences that exist for other types of minimally invasive tooth preparations for polymeric restorative materials (composite resin) are noted.

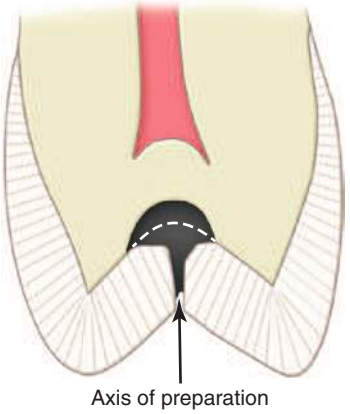
Occlusal Contact Identification and Rotary Instrument Axis Alignment

Class I, II, III, IV, and VI preparations may involve surfaces that are brought into direct occlusal contact with opposing tooth structure during function. Identification of the precise area of occlusal contact is essential so as to prevent the placement of a preparation margin (and subsequent preparation/restoration interface) where the occlusal contact occurs. Occlusal contact on the preparation/restoration interface will increase the risk of early failure of the restoration. Identification of the occlusal contact areas may be accomplished by use of articulating paper (Fig. 4.10).

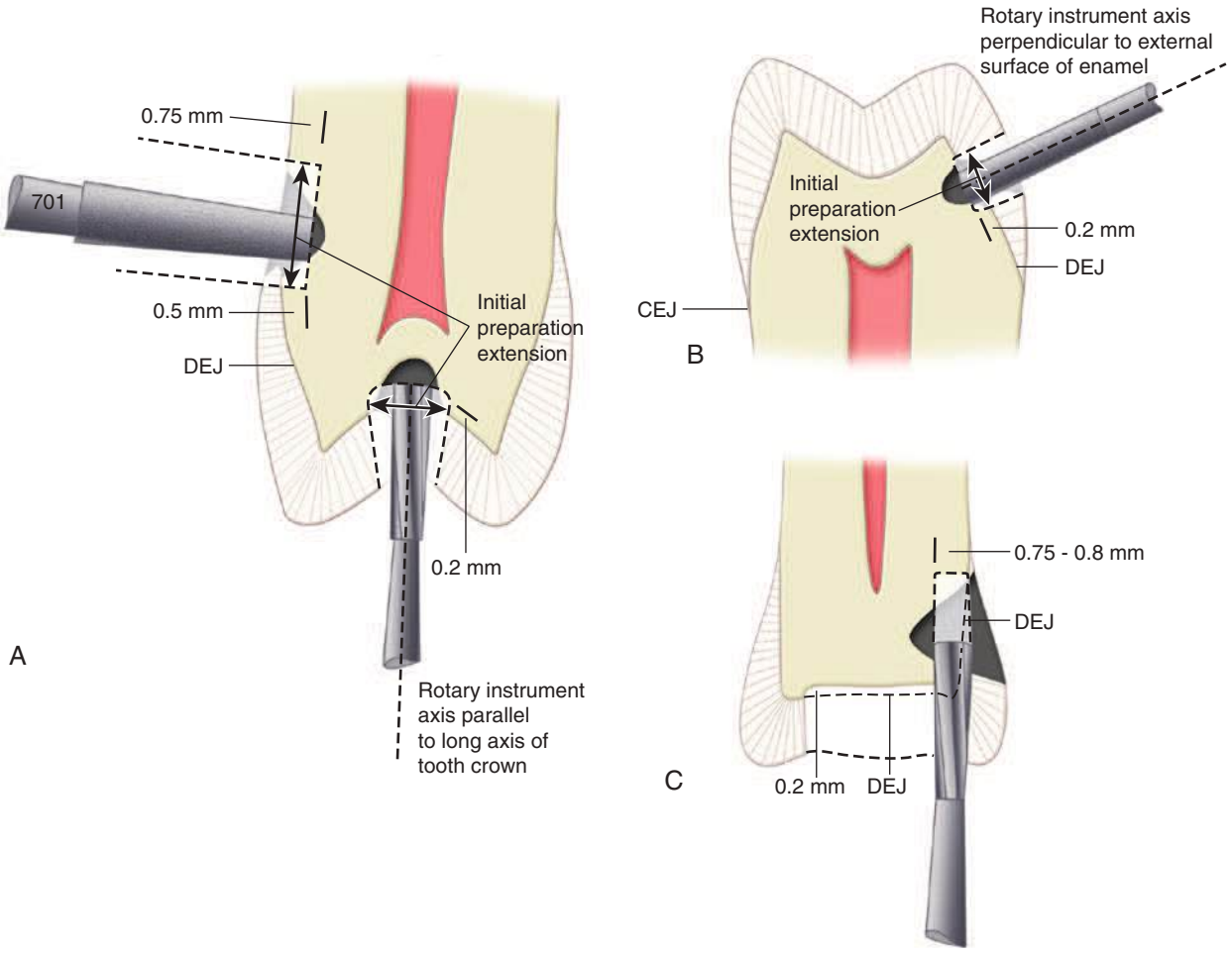
The anatomic orientation of caries lesion formation in the pit and fissure areas of posterior teeth requires alignment of the rotary instrument shank axis (through proper positioning of the handpiece) so that it is parallel with the long axis of the tooth crown *prior to* initiation of the preparation (see Online Chapter 14 for information on handpieces and rotary instruments, specifically Fig. 14.18). Correct alignment of the long axis of the shank limits the likelihood of iatrogenic removal, and thereby weakening, of adjacent healthy (occasionally referred to as "sound") coronal tooth structure. Most proximal caries lesions associated with posterior teeth also require that the shank axis be aligned parallel with the long axis of the tooth crown (Figs. 4.11 and 4.12; also see Fig. 4.2). Caries lesion formation associated with the facial or lingual surfaces of the dentition require that the shank axis be aligned perpendicular to the external surface of the tooth where the lesion is located (see Fig. 4.12B).



• **Fig. 4.10** Occlusal contact areas identified through the use of articulating paper.



• **Fig. 4.11** Diagram of a carious fissure. Note that the fissure is parallel to the long axis of the posterior tooth crown. The axis of the initial preparation into the carious fissure is aligned with the long axis of the tooth crown so as to prevent iatrogenic removal of adjacent healthy tooth structure.



• **Fig. 4.12** Initial tooth preparation stage for conventional preparations. A–C, Extensions in all directions are to healthy, mineralized ("sound") tooth structure, while maintaining a specific limited pulpal or axial depth regardless of whether end (or side) of rotary instrument is in a caries lesion or old restorative material. The dentinoenamel junction (DEJ) and the cementoenamel junction (CEJ) are indicated in B. In A, initial depth is approximately two thirds of 3-mm rotary instrument head length, or 2 mm, as related to prepared facial and lingual walls, but is half the rotary instrument (specifically the No. 245 carbide bur) head length, or 1.5 mm, as related to central fissure location.

Initial Tooth Preparation Stage: Steps 1-4

Step 1: Initial Depth and Outline Form

The first step in tooth preparation is to establish the *initial depth* and then, at that depth, extend the walls of the preparation until the junction between the enamel and supporting dentin is uncompromised (i.e., a "sound DEJ" has been reached; see Fig. 4.12). The peripheral walls determine the overall outline of the preparation, which is referred to as the *outline form*. It is essential that the outline form be visualized (i.e., mentally anticipated) as much as possible before any mechanical alteration of the tooth has begun. The initial depth of the preparation is 0.2 mm internal to the DEJ or 0.8 mm internal to the normal root surface (see Fig. 4.12). Establishment of the initial depth is always accomplished even when there is a large caries lesion (i.e., no healthy tooth structure immediately adjacent to the point of entry) or when previous restorative material is present. An exception to this initial depth of 0.2 mm internal to the DEJ is when the enamel is thin and greater depth is necessary for the strength of the restorative material that will be used. The initial preparation depth is 0.5 mm internal to the DEJ in any area where secondary retention features are being planned (see Step 7). Root-surface preparations may be shallower than 0.8 mm if the restorative material to be used does not require secondary retention features. Generally, the goals behind the establishment of initial depth and outline form on occlusal surfaces are the preservation of cuspal and marginal ridge strength (where possible). These goals are accomplished by limitation of the depth of the preparation into dentin and the minimization of faciolingual and mesiodistal extensions. Careless, iatrogenic removal of healthy dentin further compromises the diseased tooth and must be avoided. The outline form is designed, regardless of the type of tooth preparation, such that (1) all unsupported or weakened (friable) enamel is usually removed, (2) all faults are included, and (3) all preparation margins are usually placed in a position that allows inspection and finishing of the subsequent restoration margins. See Chapters 8 and 10 for exceptions to these general principles.

Black theorized that, in tooth preparations for smooth-surface caries, the initial preparation should be further extended to areas that are normally self-cleansing so as to prevent recurrence of caries around the periphery of the restoration.[1] This principle was known as *extension for prevention* and was broadened to include the extension necessary to remove remaining enamel imperfections, such as deep, noncarious fossae and grooves, on occlusal surfaces. The practice of extension for the prevention has virtually been eliminated, however, because of the relative caries immunity provided by preventive measures such as fluoride application and improved education relative to oral hygiene and diet. This change has fostered a more conservative tooth preparation philosophy. Current factors that dictate extension on smooth surfaces include (1) the extent of caries or injury and (2) the restorative material to be used. Likewise, extension for prevention to include the caries-prone areas on occlusal surfaces has been reduced by treatments that conserve tooth structure. Such treatments may include enameloplasty, application of pit-and-fissure sealant, and preventive resin or conservative composite restoration.[7]

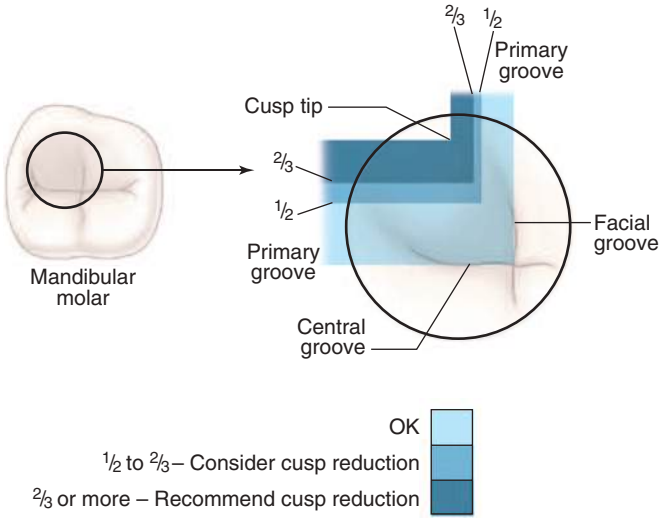
Esthetic considerations not only affect the choice of restorative material but also the design of the tooth preparation in an effort to maximize the esthetic result of the restoration. Correcting or improving occlusal relationships also may necessitate altering the tooth preparation to accommodate such changes, even when the involved tooth structure is not faulty (i.e., a cuspal form may need to be altered so as to improve occlusal relationships). Likewise, the adjacent tooth contour may dictate specific preparation extensions that enable the creation of appropriate proximal restoration form. Occasionally the tooth preparation outline for a new restoration contacts or extends slightly into a sound, existing restoration (e.g., a new MO abutting a sound DO). This approach is an acceptable practice (i.e., to have a margin of a new restoration placed into an existing, clinically acceptable restoration). Lastly, the desired cavosurface marginal configuration of the proposed restoration affects the outline form. Restorative materials that need beveled margins require tooth preparation outline form extensions that must anticipate the final cavosurface position and form that will result *after* the bevels have been placed.

Step 2: Primary Resistance Form

Primary resistance form may be defined as the shape and placement of the preparation walls (floors) that best enable the remaining tooth structure, as well as the anticipated restoration, to withstand masticatory forces primarily oriented parallel to the long axis of the tooth. Such floors may be purposefully prepared to provide a level supporting surface for the restoration, allowing a broader area for stress distribution. Primary resistance form is obtained through use of a preparation design that conserves as much healthy tooth structure as possible. Carefully controlled extension of the preparation walls allows conservation of the dentin support of adjacent cusps (and marginal ridges when possible), which helps to maintain maximum strength and therefore resistance to fracture during the cyclic loading of mastication. Transitions between the walls of the preparation (i.e., the internal line angles) are slightly rounded so as to limit stress concentration in these areas, which increases tooth resistance to fracture.[8,9] Rounding of external angles within the tooth preparation (e.g., axiopulpal line angles) limits the likelihood of stress concentration in the corresponding intaglio surface of restorative materials, which increases resistance to fracture of the restorative material. The relatively horizontal pulpal and gingival walls, prepared perpendicular to the tooth's long axis, help the restoration resist occlusal forces and limit the likelihood of tooth fracture from wedging effects caused by opposing cusps. It may be necessary to reduce cusps that no longer have sufficient dentin support and cover (or envelope) them with an adequate thickness of restorative material in order to provide resistance to fracture of the tooth and/or the restorative material. Preparation design must allow for adequate thickness of polycrystalline restorative materials to ensure adequate primary resistance to restoration fracture. The minimal occlusal thickness, for appropriate resistance to fracture, of amalgam is 1.5 to 2 mm and glass-ceramic is 2 mm. Polymeric restorative materials (e.g., composite resins) have no minimal thickness.

When developing the outline form in Class I and II preparations, the end of the cutting instrument prepares a relatively horizontal pulpal wall of uniform depth into the tooth (i.e., the pulpal wall follows the original occlusal surface contours and the DEJ, which are approximately parallel; see Fig. 4.12A and C). Similarly, in the proximal portion of Class II preparations, the end of the cutting instrument prepares a gingival wall (floor) that is approximately parallel to the occlusal surface and, thereby, relatively perpendicular to occlusal forces.

When an extensive caries lesion is present, facial or lingual extension of pulpal or gingival walls may require (1) reduction of weak cusps for coverage by the restorative material (Fig. 4.13) and/or (2) extension of the gingival floors around axial tooth line angles onto facial or lingual surfaces. These preparation modifications provide resistance to parallel and also obliquely (laterally) directed



• **Fig. 4.13** Rule for cusp reduction and coverage: If extension from a primary groove toward the cusp tip is no more than half the distance, no cusp reduction and coverage should be done; if the extension is one half to two thirds of the distance, consider cusp reduction and coverage; if the extension is more than two thirds of the distance, usually reduce the cusp and cover it with restorative material. Cusp reduction and coverage has also been referred to as "cusp capping."

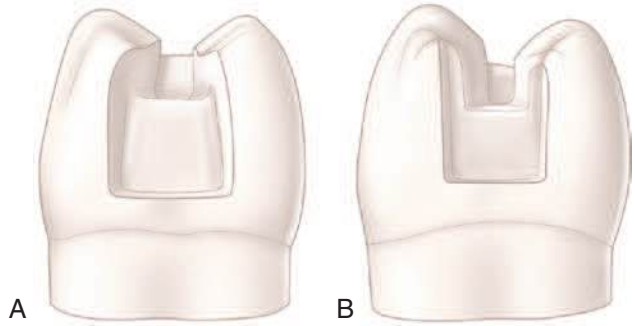
occlusal forces. The decision to reduce a cusp should be approached judiciously. The most important aspect in the evaluation of a suspicious cusp is the judgment of the amount of remaining dentin support. Cusp reduction should be *considered* when the outline form has extended half the distance from a primary groove to a cusp tip. Cusp reduction is *strongly recommended* when the outline form has extended two thirds the distance from an adjacent primary groove to the cusp tip. The exception to reducing a cusp, where extension has been two thirds from a primary groove toward the cusp tip, is when the operator judges that adequate cuspal strength (adequate dentin support) remains. Reduction of cusps occurs as early as possible in the preparation process so as to improve access and visibility for the operator.

Special consideration is given to teeth that have lost an excessive amount of dentin support in the central area of the tooth secondary to endodontic procedures. Adjacent cusps may be considerably compromised and, as such, may need to be reduced, enveloped, and covered with restorative material to prevent subsequent catastrophic fracture when under occlusal load.[10,11] In general, the greater the occlusal load, the greater is the potential for future fracture of the tooth and/or restoration. The further posterior the tooth is, the greater is the effective occlusal load because of its closer location to the hinge axis (i.e., the temporomandibular joint [TMJ]).

Weakened, friable tooth structure always should be removed at this step in the preparation. Unsupported but not friable enamel may be left for esthetic reasons in anterior teeth where stresses are minimal and a bonded composite restoration is anticipated.

Step 3: Primary Retention Form

Primary retention form is the shape or form of the preparation that prevents displacement or removal of the restoration by tipping or lifting forces. In many respects, retention form and resistance form are accomplished at the same time (Fig. 4.14). The retention form



• **Fig. 4.14** Basic primary retention form in Class II tooth preparations for amalgam (A) with vertical external walls of proximal and occlusal portions converging occlusally and for inlay (B) with similar walls slightly diverging occlusally.

developed during initial tooth preparation may be adequate to retain the restorative material in the tooth.

The design of preparation primary retention form is directly related to the retention needs of the anticipated restorative material. Amalgam restoration of a Class I or II preparation is retained by developing external tooth walls that converge occlusally (see Fig. 4.14A). In this way, when the amalgam is placed in the preparation and hardens, it cannot be dislodged. However, excessive occlusal convergence of the external walls will result in unsupported enamel rods at the cavosurface margin and must be avoided. The external walls of Class III and V preparations diverge so as to provide strong enamel margins (see Figs. 1.12B and 4.12A and B). Retention of amalgam in these areas requires the creation of secondary features (coves or grooves) in the dentinal walls that serve to retain the restoration (see Step 7).

Composite restorations are primarily retained in the tooth by micromechanical and, depending on the adhesive, chemical bonding that is established between the restoration and the tooth structure. In such restorations, the preparation surface of the enamel and dentin are etched (demineralized) by creation of acidic conditions and then infiltrated with resin-based adhesive materials before placement of the composite.

Cast-metal intracoronal restorations, referred to as inlay restorations, rely on diverging vertical walls that are almost parallel and a luting cement to provide retention of the casting in the tooth (see Online Chapter 18). During the initial tooth preparation, the preparation walls are designed not only to provide for draw (for the casting to be placed into the tooth) but also to provide for an appropriate small angle of divergence (2–5 degrees per wall) from the line of draw (to enable retention of the luted restoration). The amount of divergence required depends on the length of the prepared walls: The greater the vertical height of the walls, the more divergence is permitted and recommended, but within the range described. The path of draw is usually designed to be perpendicular to the horizontal features of the preparation (see Fig. 4.14B).

Step 4: Convenience Form

Convenience form is the shape or form that provides adequate observation, accessibility, and ease in the preparation and restoration of the tooth. Severe caries destruction may necessitate the extension of distal, mesial, facial, or lingual walls so as to gain adequate access to deeper areas of the preparation. Extension of the proximal walls, so as to obtain clearance with an adjacent proximal surface, may afford better access for the finishing of preparation walls, the placement of the matrix, and the finishing of the restoration margins.

The arbitrary extension of facial margins on anterior teeth is usually contraindicated, however, for esthetic reasons. Clearance with the adjacent proximal surface is mandatory for glass-ceramic and cast gold restorations because of the need to finish the preparation walls, make an accurate impression of the prepared tooth, and accomplish insertion/finishing. Preparation extensions to increase the convenience of various procedures are always accomplished in light of the goal of conserving as much healthy tooth structure as possible.

Final Stage of Tooth Preparation: Steps 5-9

When the design of external wall orientation and position has fulfilled the objectives of initial tooth preparation, the preparation is carefully inspected for other needs. The preparation may be complete after the initial tooth preparation stage when the caries lesion (or other defect) is minimal. In this case the preparation will then only require (1) desensitization of the prepared dentin walls for amalgam or (2) modification of the surface of the enamel and dentin so as to create an adhesive interface for composite.

Step 5: Removal of Defective Restorative Material and/or Soft Dentin

Once the caries lesion has been fully exposed (via the initial preparation), careful pulpal and/or axial extension is accomplished so as to *remove defective restorative material and/or soft dentin* as indicated (see Chapter 2). Old restorative material may remain on the pulpal or axial walls after initial tooth preparation. This material should be removed if any of the following conditions are present: (1) the old material may negatively affect the esthetic result of the new restoration (i.e., old amalgam material left under a new composite restoration), (2) radiographic evidence indicates caries lesion development under the old material and clinical evaluation confirms the radiographic interpretation, (3) the tooth pulp was symptomatic preoperatively, (4) the dentin along the periphery of the remaining old restorative material is soft, or (5) retention of the existing material is compromised and the material is easily dislodged. If none of these conditions is present, it is acceptable to leave the remaining old restorative material to serve as a base, rather than risk unnecessary excavation in close proximity to the pulp, which may result in pulpal irritation or exposure. Removal of remaining old restorative material, when indicated, may be accomplished using sharp rotary instruments and light intermittent pressure with or without water irrigation/cooling. Water spray (along with high-volume evacuation) is used when removing old amalgam material to reduce exposure to mercury vapor.

In preparations that remain primarily in enamel, isolated faulty areas (remnants of diseased enamel fissure or pit) on the pulpal wall may require additional minimal extensions. When a pulpal or axial wall has been established at the proper initial tooth preparation position, and a small amount of carious tissue remains, only this tissue should be removed, leaving a rounded, concave area in the wall. The level or position of the wall peripheral to the excavation should not be altered.

Clinical decisions that guide carious tissue removal are based on the relative tactile hardness (firmness) of the dentin associated with the caries lesion. Carious dentin that has had some mineral loss, but not to the point of collagen exposure, is not as clinically hard as normal dentin and is referred to as *firm dentin.* Firm dentin, if isolated from the oral environment by some type of restoration, will remineralize and therefore should not be removed.[12] If further demineralization causes exposure of the collagen matrix, subsequent enzymatic degradation of the matrix begins as a result of the activity of host matrix metalloproteinases (MMPs) and cysteine cathepsins. Denaturation of the collagen, by host proteolytic enzymes, allows subsequent collagen degradation (of the denatured collagen) by bacterial proteases. Bacterial proteases are not able to degrade intact, native collagen. The process of denaturation and degradation changes the three-dimensional structure of the collagen such that remineralization is no longer possible. Carious tissue that has been demineralized and structurally damaged to this level feels tactilely soft and is therefore referred to as *soft dentin*. Soft dentin (previously referred to as *infected dentin* because of high numbers of bacteria) no longer retains the physical properties necessary to survive in the rigors of the oral environment and must be removed except in the deepest areas of the preparation where removal would result in exposure of an asymptomatic, vital pulp (see Chapter 2, selective caries removal [SCR] protocol).

The use of color alone to determine how much dentin to remove is unreliable. For example, an area of dentin that has experienced episodes of demineralization and remineralization often clinically appears discolored, compared with normal dentin, yet may be firm to tactile exploration and should not be removed. Alternatively, acute (rapid) caries often manifests itself entirely within the normal range of color for dentin and is tactilely soft. The soft dentin requires removal except as indicated (see Chapter 2). Additionally, when the conditions in the lesion have allowed remineralization to occur, the dentin may be distinctly discolored or "stained." In this case host defenses not only have enabled remineralization of the dentin, which is often clinically comparable in firmness (hardness) with surrounding normal dentin, but also have, for the most part, successfully filled in the dentinal tubules with mineral. This resultant hypermineralized state of the dentin (mineralization above that which is found in normal dentin as the lumens of the dentinal tubules are filled with mineral in addition to the normal mineralization of the intertubular dentin) is referred to as *sclerotic dentin*. Sclerotic dentin should not be removed.

Removal of carious dentin is accomplished with awareness of the ability of the vital pulp to effect remineralization of dentin when the matrix (collagen) has not been denatured. Therefore every effort should be made to limit further pulpal irritation and limit the likelihood of pulpal involvement during the caries removal process. The use of sharp spoon excavators and sharp rotary instruments, with intermittent light pressure, may help limit pulpal irritation. Removal of carious tissue in a *moderate lesion* (i.e., a lesion that has not reached the inner one third of dentin) has a low risk of pulpal involvement. The pulpal and axial caries removal of a moderate lesion should therefore extend to where the dentin is firm to tactile sense (i.e., extend to *firm dentin*). Removal of a carious tissue in an *advanced lesion* (i.e., a lesion that has reached the inner one third of dentin) has a higher risk of pulpal involvement. The pulpal and axial caries removal of an advanced lesion should therefore extend to approximately 1 mm from the pulp with the recognition that dentin in this deep region may still be soft *(soft dentin)* to tactile sense. Carious dentin in more peripheral areas is removed until the dentin is firm.

In dentin, as the caries lesion progresses, a zone of demineralization precedes the invasion of, or infection by, bacteria. It is currently impossible to clinically identify the specific depth of the bacterial invasion. Previous notions of dentin excavation for the purpose of complete removal of all bacteria have resulted in exposure of pulp tissue that was not irreversibly inflamed, leading to overtreatment and increased frequency of adverse outcomes. It

is not necessary that all dentin invaded by bacteria be removed. In moderate caries lesions, removal of the masses of bacteria and subsequent sealing of the preparation by a restoration at best destroy those comparatively few remaining microorganisms and at worst reduce them to inactivity or dormancy.[13] Even in advanced caries lesions, in which actual invasion of the pulp may have occurred, the recovery of the pulp requires only that a favorable balance be established between the virulence of the bacteria and the resistance of the host. This balance is best accomplished by utilization of the selective caries removal protocol (see Chapter 2). Caries removal in advanced lesions usually is immediately followed by efforts to afford protection to the pulp tissue adjacent to the deepest area of the preparation.

Step 6: Pulp Protection

Deep dentin is very porous and susceptible to desiccation. The thin remaining wall of dentin provides little protection from (1) heat generated by rotary instruments during subsequent steps, (2) noxious ingredients of various restorative materials, (3) thermal changes conducted through restorative materials, (4) forces transmitted through materials to the dentin, (5) galvanic shock, and (6) the ingress of bacteria and/or noxious bacterial toxins through microleakage.[14,15] Deep dentin also is a very poor substrate for subsequent bonding procedures. Therefore efforts to cover deep dentin, to limit dentinal tubular fluid flow, and to create a protective thermal/physical barrier are warranted. Various materials that have been utilized to establish this protective barrier include suspensions or dispersions of zinc oxide, calcium hydroxide, or resin-modified glass ionomer (RMGI) that are applied to the tooth surface.[15] These materials are referred to as *liners* when used in a relatively thin film.[15] The term *base* is used to describe the placement of materials, used in thicker dimensions, beneath permanent restorations to provide for mechanical, chemical, and thermal protection of the pulp. Examples of bases include zinc phosphate, zinc oxide–eugenol, polycarboxylate, and most commonly, some type of glass ionomer material (usually a RMGI). Generally, it is desirable to have approximately a 2-mm dimension of bulk between the pulp and a metallic restorative material. This bulk may include remaining dentin, liner, or base. Composite resin materials, which are thermal insulators, do not require the same bulk of material (dentin + liner/base) between the restoration and the pulp. Although the placement of liners and bases is not a step in tooth preparation, in the strict sense of the term, these serve to create an effective barrier over the deep pulpal/axial dentin prior to receiving the final restorative material.

If the removal of soft dentin does not extend deeper than 1 to 2 mm from the initially prepared pulpal or axial wall, usually no liner is indicated. If the excavation extends to within 0.5 mm of the pulp, a liner usually is selected to cover the deepest area of the dentin. The stimulation of reparative dentin (indirect pulp cap procedure) in this deep area with a calcium hydroxide ($CaOH_2$) liner may be effective.[16] Current evidence suggests that the actual type of material used for the liner is not as important as the overall effective sealing of the dentin with the liner (or base) and subsequent restoration.[17]

Zinc oxide–eugenol and calcium hydroxide liners (chemosetting types that harden) in thicknesses of approximately 0.5 mm or greater have adequate strength to resist condensation forces of amalgam and provide insulation against thermal extremes.[18] $CaOH_2$ liners must always be covered with a RMGI when used under amalgam restorations to prevent dissolution of the liner over time. In addition, $CaOH_2$ liners should be covered by a RMGI to protect the liner from dissolution from the phosphoric acid etchant used prior to composite placement.[14,19] Protection of the $CaOH_2$ liner with an RMGI base also prevents inadvertent displacement of the liner during subsequent procedural steps.

Very deep excavations may contain microscopic pulpal exposures that are not visible to the naked eye. Hemorrhage is the usual evidence of a vital pulp exposure, but with microscopic exposures, such evidence may be lacking. Nevertheless, these exposures may be large enough to allow direct pulpal access for bacteria or other restorative materials. The ability of a hard-setting $CaOH_2$ material to stimulate the formation of reparative dentin when in contact with pulpal tissue makes it the usual material of choice for application to very deep excavations and known pulpal exposures (direct pulp cap procedures).[16] Alternatively, mineral trioxide aggregate (MTA) liners have been found to be effective for direct pulp capping.[17,20] Liners and bases in exposure areas should be applied without pressure.

Usually, a RMGI is used for "base" needs. These materials effectively bond to tooth structure, release fluoride, and have sufficient strength. Placement and contouring of RMGI materials is readily accomplished. Mechanical retentive preparation features are not typically required for RMGI because of their chemical bond to the mineral phase of tooth structure. These materials are excellent for use under amalgam, gold, ceramic, and composite restorations. A base should never overfill the preparation and, thereby, compromise the minimum required thickness of the final restoration.

A liner may be utilized to seal off the dentin immediately adjacent to the pulp when traumatic fracture has occurred. The desired pulpal effects may include sedation and stimulation, the latter resulting in reparative dentin formation. The specific pulpal response desired dictates the choice of liner material. An appropriate text focused on the emergency management of dental trauma should be consulted for current treatment strategies.

Step 7: Secondary Retention and Resistance Forms

Placement of secondary retention and resistance forms, as part of a preparation, follows management of the caries lesion and any indicated pulpal protection. Each anatomically distinct area requiring restoration must be independently retentive. When the external walls of the preparation converge toward each other, as they approach the external surface of the tooth, then no additional or "secondary" retention is required. However, correctly oriented external walls (i.e., walls that have proper dentinal support of the enamel) may diverge as they approach the external surface of the tooth. For example, preparation of a proximal caries lesion on a posterior tooth will frequently result in facial, lingual, and gingival walls that diverge proximally. Preparation walls that diverge will not physically retain a restoration that is not bonded in place. Diverging walls will not resist forces that have the potential to result in the dislodgement of a restoration. Therefore it may become necessary to strategically modify internal aspects of the preparation so as to mechanically retain the restoration.

Because many preparation features that improve retention form also improve resistance form, and the reverse is true, they are presented together. The *secondary retention and resistance forms* are of two types: (1) mechanical preparation features and (2) treatments of the preparation walls with etching, priming, and adhesive materials. The second type is not really considered a part of tooth preparation but, rather, the first step for the insertion of the restorative material. Regardless, some general comments are presented about such treatments.

Mechanical Features

A variety of mechanical alterations to the preparation enhance retention form. These alterations require additional selective removal of healthy tooth structure.

Retention Grooves and Coves. Vertically oriented grooves associated with the facial and lingual aspects of a proximal preparation are used to provide additional retention for the proximal portions of some Class II amalgam restorations. Horizontally oriented retention grooves are prepared in most Class III and V preparations for amalgam and in some root-surface tooth preparations for amalgam and composite resin. Small retentive indentions, referred to as "coves," are utilized for retention in the incisal areas of Class III amalgams.

Historically, retention grooves in Class II preparations for amalgam restorations were recommended to increase retention of the proximal portion against movement secondary to creep. Also, it was thought that retention grooves may increase the resistance form of the restoration against fracture at the junction of the proximal and occlusal portions. *In vivo* studies do not substantiate the necessity of these grooves in proximocclusal preparations with occlusal dovetail outline forms or in MOD preparations.[4] They are recommended, however, for extensive tooth preparations for amalgam involving wide faciolingual proximal boxes resulting in notable proximal wall divergence, cusp reduction procedures, or both. Therefore mastery of the techniques of optimal groove design and placement is indicated.

Preparation Extensions. Additional retention of the restorative material may be obtained by arbitrarily extending the preparation for molars onto the facial or lingual surface to include a facial or lingual groove. Such an extension, when performed for cast-metal restorations, results in additional vertical (almost parallel) walls for retention. This preparation design may also enhance the resistance form of the remaining tooth by enveloping and contributing reinforcement.

Skirts. Skirts are preparation features used in cast gold restorations that extend the preparation around some, if not all, of the line angles of the tooth. When properly prepared, skirts provide additional, opposing vertical walls that increase retention of the restoration. The placement of skirts also enables increased resistance to fracture by allowing the envelopment of the remaining compromised tooth structure with the restorative material.

Beveled Enamel Margins

Some cast-metal and composite preparations include beveled marginal configurations. The bevels for cast-metal restorations are used primarily to afford a better junctional relationship between the metal and the tooth. Additionally, retention form may be slightly improved when opposing bevels are present. Enamel margins of some composite restorations may utilize a beveled or flared (>90 degrees) configuration so as to increase the retention form of the preparation by increasing the area of enamel available for bonding.

Steps, Amalgam Pins, Slots, and Pins

When the need for increased retention form for amalgam is unusually great (i.e., there is limited remaining tooth structure available to help retain the restoration), additional secondary features may be incorporated into the preparation. Careful orientation of remaining horizontal and vertical walls during tooth preparation results in "steps" that increase retention and resistance form of the restoration. Independently retentive holes with parallel walls (for "amalgam pins") and/or horizontal slots (with internal converging walls) may be effectively used when there is moderate vertical loss of tooth structure. Severe vertical loss of structure associated with the line angles of the tooth may require the placement of metal pins. These pins are anchored in remaining sound dentin, protrude vertically above the remaining tooth structure, are subsequently encased during placement of the restorative material, and thereby enable retention and resistance form. Features that enhance the retention form of a preparation also enhance the resistance form (e.g., slots or pins placed in a manner such that, upon completion of the restoration, the structural integrity of the restoration enhances the structural integrity of the remaining tooth structure).

Use of Adhesives to Increase Retention and Resistance

Superficial demineralization of preparation walls and subsequent infiltration of the altered surface with resin-based adhesives allows for increased retention and resistance of restorations. The structural makeup of enamel allows the creation of a microscopically roughened mineral surface when superficially demineralized by acidic conditions. All adhesive systems have some means by which to effect the necessary demineralization. This essential, initial step is then followed by infiltration of the roughened surface with resin-based adhesive materials. Demineralization of the exposed dentin surface results in exposure of the dentin matrix (collagen), which may then be infiltrated with adhesive resin materials. Restorative materials (composite, glass-ceramic) may then be attached to this adhesive layer through material-specific mechanisms resulting in increased retention of the "bonded" restoration. The use of bonding systems with intracoronal restorations, while enhancing retention, does not increase the resistance form of the remaining tooth structure over the long term.

Retention of indirect restorations may be enhanced by the material used for cementation. Although not considered part of the tooth preparation, the cementation procedure does affect the retention of these restorations, and some cementing materials require pretreatment of the dentin, resulting in varying degrees of micromechanical bonding. Adhesive bonding of etchable glass-ceramic materials to enamel and dentin increase their resistance to fracture development when under occlusal load.

Step 8: External Wall Finishing

Finishing the external preparation walls is the further development, when indicated, of a specific design (e.g., degree of smoothness or roughness, the placement of a bevel) immediately adjacent to or including the cavosurface margin such that the anticipated restorative material has the greatest likelihood of clinical success. Proper finishing of the external walls allows the creation of an optimal marginal junction between the restorative material and the tooth structure. When this occurs there is a smooth transition across the marginal junction and both tooth and restorative material have maximal strength. Awareness of the point of contact of the opposing functional cusp is essential as occlusal contact directly on the marginal interface will result in early fatigue and failure of the margin. When the defect results in a preparation outline form that places the marginal interface at the point of contact, then the final position of the preparation outline is modified slightly so that the marginal junction is away from the occlusal contact (review section Occlusal Contact Identification and Rotary Instrument Axis Alignment).

It is appropriate, for clinical practicality, to consider that enamel rods are oriented perpendicular to the external tooth surface. Enamel walls that form a 90-degree angle with the cavosurface may be considered to have dentinal support and to be strong (see Figs. 4.1 and 4.2C). Enamel rods that do not run uninterrupted from

the preparation margin to dentin tend to split off, leaving a V-shaped ditch along the cavosurface margin area of the restoration. Enamel rods incline slightly apically in the gingival third of the tooth crown and preparation design in this area should be modified so as to ensure strong enamel margins (Fig. 4.15).

When a preparation has extended onto the root surface (i.e., no enamel present), the root-surface cavosurface angle should be either 90 degrees (for amalgam, composite, or ceramic restorations) or beveled (for intracoronal cast-metal restorations). The 90-degree root-surface margin provides a butt joint relationship between the restorative material and the dentin (with overlying cementum) preparation wall, a configuration that provides appropriate strength to both.

An acute, abrupt change in a preparation wall outline form increases the difficulty of optimal adaptation of the restorative material. The outline form of all preparation walls should have smooth curves or straight lines. The line angle that forms where two walls meet, regardless of whether it is acute or obtuse, should be slightly curved ("softened") (Fig. 4.16).

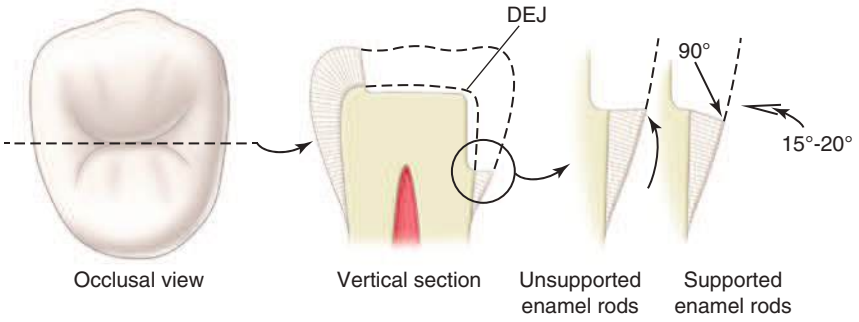
The design of the cavosurface angle depends on the restorative material being used. Because of the low edge strength of amalgam and glass-ceramic, a 90-degree cavosurface angle produces maximal strength for these materials. Beveling a cavosurface margin would result in a thin, fracture-prone amalgam or ceramic margin and is contraindicated. On occlusal surfaces for Class I and II amalgam restorations, the incline planes of the cusp and the converging walls (for retentive purposes) of the preparation approximate the desirable 90-degree butt joint junction, even though the actual occlusal enamel margin may be greater than 90 degrees (see Figs. 4.1, 4.2, 4.8, and 4.12).

Beveling the external walls is a preparation technique used for some materials, such as intracoronal cast gold and composite restorations. As previously noted, beveling will result in the strongest type of enamel margin. When a bevel is used for indirect cast gold restorations, it may allow a better marginal seal in slightly undersized castings and may assist in the adaptation of castings that fail to seat by a small amount. In addition, a thin gold margin is more readily burnished and adapted to the preparation margin.

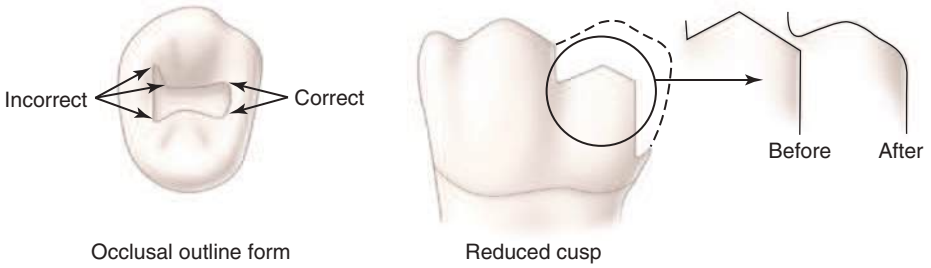
Beveling enamel margins in composite preparations is indicated primarily for large restorations that have increased retention needs and insufficient amount of prepared enamel for proper retention. The use of a beveled marginal form increases the surface area available for bonding, which increases the retention form of the preparation. Diamond instruments are utilized to create the bevel so as to maximize the surface area for bonding. The use of a beveled marginal form is useful for inclusion of minor surface defects just adjacent to the cavosurface margin as well as affords enhanced marginal sealing. The esthetic quality of composite restorations of anterior teeth may be improved by use of a bevel to create an area of gradual increase in composite thickness from the margin to the bulk of the restoration. Hand instruments such as enamel hatchets and margin trimmers may be used in planing enamel walls, cleaving off unsupported enamel, and establishing enamel bevels.

Step 9: Final Procedures: Debridement and Inspection

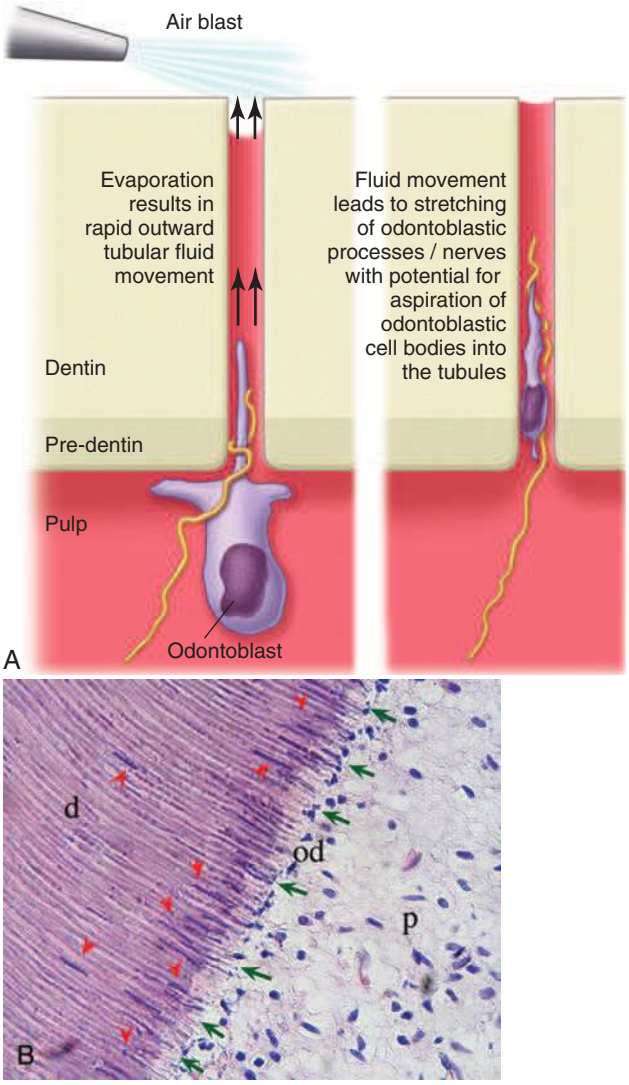
Debridement (cleaning) of the tooth preparation involves use of the air/water syringe to remove visible debris with water and then excess moisture with a few light bursts of air. In some instances, debris clings to walls and angles despite the aforementioned efforts, and it may be necessary to loosen this material with an explorer or small cotton pellet. It is important not to dehydrate the tooth by overuse of air as this may damage the odontoblasts associated with the desiccated tubules (Fig. 4.17). Complete debridement allows careful *inspection* of the preparation so as to ensure adherence to all principles of preparation design. Any final changes may then be accomplished, as indicated, followed by steps to disinfect the preparation.



• **Fig. 4.15** Vertical section of Class II tooth preparation. Gingival floor enamel (and margin) is unsupported by dentin and friable unless removed.



• **Fig. 4.16** The junctions of enamel walls (and respective margins) should be slightly rounded, whether obtuse or acute.



• **Fig. 4.17** A, Excessive drying (desiccation) of tooth preparations may cause odontoblasts to be aspirated into dentinal tubules. B, Nuclei are seen as dark rods in dentinal tubules. Red arrowheads indicate the nuclei of the aspirated odontoblasts. Green arrows indicate location of the odontoblasts prior to them being drawn into the tubules from outward dentinal tubular fluid flow. *d*, dentin; *od*, odontoblasts; *p*, pulp. (B, From Mitsiadis TA, De Bari C, About I: Apoptosis in development and repair-related human tooth remodeling: A view from the inside, *Exp Cell Res* 314(4):869–877, 2008.)

Additional Concepts in Tooth Preparation

New techniques advocated for the restoration of teeth should be assessed on the basis of the fundamentals of tooth preparation presented in this chapter. All preparations in stress-bearing areas, once completed, should ensure healthy dentinal support of remaining enamel.

Tunnel Tooth Preparations for Amalgam, Composite Resin, and Glass Ionomers

In an effort to be conservative in the removal of tooth structure, some investigators advocate a "tunnel" tooth preparation. This preparation joins an occlusal lesion with a proximal lesion by means of a prepared tunnel under the involved marginal ridge. The objective of this approach is to remove the caries lesion and leave the marginal ridge essentially intact. However, the development of appropriately formed preparation walls and the excavation of the caries lesion may be compromised by lack of access and visibility. Preservation of the marginal ridge in a strong state is questionable, especially since the dentinal support (essential for enamel durability under occlusal loading) of the marginal ridge is no longer present or is compromised. Generally teeth that have been treated with tunnel preparations do not perform as well as those treated with preparations that remove the marginal ridge over the proximal lesion so as to gain access to the proximal caries lesion. Most currently published clinical trials focus on the use of glass ionomer materials to restore tunnel preparations and have found these materials to be inadequate for use as definitive, long-term restorations. It is currently unknown whether use of adhesively retained composite resin materials will result in better long-term clinical performance. This technique remains controversial and is not supported in this textbook.[12,21]

Adhesive Amalgam Restorations

In vitro research studies suggest that the use of adhesive systems may enhance resistance and retention forms of teeth with compound and complex amalgam preparations/restorations.[22,23] These techniques mechanically bond the amalgam material to tooth structure in the hope that this will increase the overall strength of the remaining tooth structure and improve the overall performance of the restoration. Although proposed bonding techniques vary, the essential procedure is to prepare the tooth in a fashion similar to that typical for amalgam and then utilize an adhesive to bond the amalgam restoration into the preparation. In addition, weakened remaining tooth structure is retained and bonded to the amalgam instead of reduced and covered with amalgam. The preparation walls, including the walls of compromised areas (where there is inadequate dentin support), are treated with specific adhesive lining materials that mechanically bond to the tooth and the amalgam. The amalgam is condensed into this adhesive material before polymerization, and a mechanical bond develops between the amalgam and adhesive. It has been suggested that this technique may limit the likelihood of the development of postoperative sensitivity, staining of the dental structure, secondary caries, fracture of the tooth, or partial/total loss of the restoration. However, no clinical improvement over normal, routine steps utilized in amalgam restoration has been demonstrated.[24-28] Therefore this book does not promote the use of bonded amalgams.

Preparation Treatments to Enhance Restoration

Disinfection, Desensitization, Stabilization

Disinfection of the preparation prior to insertion of the restorative material may be considered. Chlorhexidine (2 weight percent [wt%]) solutions have been successfully used in preparations for disinfection purposes. Disinfection procedures should not be considered absolutely essential. Investigators have verified the presence of bacteria in the dentinal tubules within the preparation walls. While it is true that the dentinal tubule lumens, which vary from 1 to

4 μm in diameter depending on the depth of the preparation, present a pathway for the entrance of bacteria, this fact in no way indicates that caries progression or restoration failure will result. The number of bacteria in the dentinal tubules is relatively small compared with the numerous microorganisms found in the superficial caries lesion. The caries lesion will not progress if the defect is correctly restored.[12]

Even when surface disinfection of the preparation has been attempted, it is doubtful that potential benefits will continue for any appreciable length of time because of the difference between the thermal coefficients of expansion of the tooth and restorative materials. Although differing in amounts, marginal leakage has been shown for current restorative materials.[18,29,30] Caries is unlikely to develop in association with marginal gaps that are less than 250 μm.[29] Limited protection from further carious activity may be afforded by some restorative materials.[31] The germicidal or protective effect may be from the fluoride content of some tooth-colored restorative materials or from the deposition of corrosion products at the interface between the preparation wall and an amalgam restoration.

The natural defense mechanisms of the tooth, which result in the mineralization of the dentinal tubules under a caries lesion, help limit the potential for invasion of any remaining bacteria. However, this natural occlusion of the dentinal tubules only will occur beneath a slowly progressing caries lesion. The precipitation of mineral in the dentinal tubules beneath a caries lesion (giving it a transparent appearance) creates a physical barrier to bacterial ingress. In addition to this host-defense mechanism, the presence of reparative dentin deposited as a result of pulpal insult constitutes a significant deterrent to bacterial invasion. Bacteria may transition into dormancy as the result of the more sealed environment of a restored tooth. When mineral occlusion of the dentinal tubules has not occurred, there is increased risk of pulpal sensitivity after the restoration has been placed. Therefore routine use of medicaments to occlude the dentinal tubules (i.e. "desensitizers") is recommended in the finalized preparation.

Desensitization may be accomplished by taking steps to limit rapid fluid movement in the dentinal tubules. Occlusion of the dentinal tubules limits the potential for rapid tubular fluid movement. Some desensitizers not only are effective disinfectants but also are able to occlude ("plug") the dentinal tubules by cross-linking and precipitating the proteins in the dentinal tubule fluid.[32-35] Preparations designed for amalgam restoration should be desensitized with a solution that contains 5% glutaraldehyde and 35% 2-hydroxyethyl methacrylate (HEMA) before amalgam placement.[36] The use of this type of desensitizer allows prevention of rapid fluid movement associated with osmotic gradients and temperature gradients. Osmotic gradients cause a rapid, transitory increase in the amount of marginal microleakage associated with recently placed amalgam restorations, which may result in sensitivity. Temperature gradients (amalgam is an excellent thermal conductor) also may result in rapid fluid movement in the dentinal tubules secondary to exposure to temperature extremes. Alternatively, there are some who advocate the use of an adhesive on the prepared tooth structure so as to limit rapid fluid movement by "sealing" the dentin before amalgam placement and in this way limit the potential for postoperative sensitivity.[37]

Composite restorations require some treatment of the preparation before insertion of the restorative material, which may primarily be considered as part of the restoration procedure (see Chapter 8); however, some discussion is appropriate at this point. Preparation wall treatment for composites includes the etching (surface demineralization) of enamel and dentin followed by placement of a resin-based adhesive. The smear layer that forms on wall surfaces during preparation is either altered or removed from the enamel and dentin during the etching process. Placement of the adhesive will allow subsequent formation of strong, durable mechanical bond between the etched enamel and the composite. In dentin, a hybrid layer is formed, which is characterized by an intermingling of the resin adhesive with exposed collagen fibrils of the intertubular dentin. This initially creates a strong mechanical bond between the composite and dentin. However, it has been identified that the bond to dentin deteriorates over time as a result of hydrolysis of the adhesive resin component of the hybrid layer and proteolytic degradation of the collagen component of the hybrid layer.[38] Therefore *stabilization* of the exposed collagen may be appropriate as an initial step in the restorative sequence.

Ongoing dental research has sought to optimize the long-term stability of the hybrid layer. For example, *in vivo* studies have shown that chlorhexidine (2 wt% solution) application to etched dentin is able to limit the activity of local collagenolytic enzymes (matrix metalloproteinases, or MMPs), which are able to degrade the exposed collagen matrix. Limiting the activity of the MMPs may help stabilize the hybrid layer, at least in Class I preparations for the short term.[39] Chlorhexidine is able to inactive MMPs that are exposed as a result of the etching process. However, additional new MMPs, being generated by ongoing odontoblastic activity, will remain active and able to degrade collagen that was demineralized but not adequately infiltrated with adhesive resin. Long-term hybrid layer stability, as a result of chlorhexidine use, has not been demonstrated. These findings, as well as the decision to incorporate chlorhexidine or other dentin protease inhibitors as an initial restorative step for hybrid layer stabilization, are to be considered in light of clinical studies that reveal the clinical performance of composite systems that did not use chlorhexidine is comparable with that of amalgam in patients who are low caries risk.[40] However, it has been found that, in high caries risk patients, composite restorations do not perform as well as amalgam restorations.[40,41] Therefore there may be advantages to the use of agents that stabilize and increase hybrid layer resistance to proteolytic activity as a first step of the restoration sequence.

The use of a 5% glutaraldehyde/35% HEMA solution theoretically may be used immediately after etching and before priming of the dentin for the following reasons: (1) to occlude dentinal tubules and, thereby, limit tubular fluid contamination during hybrid layer formation, (2) to cross-link the acid-exposed intertubular collagen so as to render it resistant to proteolytic degradation, and (3) to cross-link and inactivate noncollagenous proteins that are able to degrade collagen (MMPs and cathepsins). Recent *in vitro* evidence is in support of this theory.[35] However, potential cytotoxic effects of free glutaraldehyde and HEMA (i.e., not involved in the protein cross-linking and tubular occlusion) raise legitimate patient safety concerns. Removal of excess glutaraldehyde and HEMA by rinsing with water may significantly reduce any risk. The outward flow of dentinal tubular fluid may also tend to limit the potential for free glutaraldehyde and/or HEMA to diffuse toward and negatively impact pulpal tissue. Furthermore, it is necessary to recall that pulpal and axial dentin in an advanced lesion (see Step 5 above) has been damaged by the caries process and any bond to this deep dentin is compromised. Therefore deep dentin areas of the advanced lesion should be

covered (i.e., sealed off) with a RMGI *prior* to any attempt at demineralization (either by total-etch or self-etch systems) of more peripheral dentin that might be followed by efforts (such as use of a 5% glutaraldehyde/35% HEMA solution) to stabilize and increase hybrid layer resistance to proteolytic activity as a part of the restorative sequence. Placement of the RMGI may theoretically limit the ability of any free glutaraldehyde or HEMA to gain access to tubules in closest proximity to the pulp. Additional testing to validate the safety and efficacy of this stabilization technique is indicated.

Summary

This chapter has addressed the principles of tooth preparation. Every effort should be made to conserve and protect remaining healthy natural tooth structure during the various steps of preparation. Tooth preparation is guided through careful consideration of the implications of many factors. Preparation design is strategically implemented so as to provide the subsequent restoration with an optimal chance of clinical success.

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